D1 - Team Charter and Work Plan

MIN E 413 Term Project - Long-Term Stockpile Reconstruction and Block Modelling

> **Before submission:** enter real names in `config/team.yaml` (they replace every `[Member n]` in the built documents), confirm the roles at the first meeting, and have every member sign Section 9. The meeting log and contribution record (Section 8) must reflect what *actually* happened. Do not pre-fill them.

1. Team members and initial roles

The team has five members. Each member owns one technical work package. Every member also reviews another member's work package (Section 6), so that each person can explain the whole method at the technical defence (course outline, Section 7.3).

Member	Primary role (owner)	Technical work package	Reviewer of	Rotating duty
[Member 1]	Project lead and integrator	Overall workflow, final report (D6), presentation (D7)	WP5	Meeting chair (weeks 1-3)
[Member 2]	Data and QA/QC lead	Task 1 data investigation, Task 2 growth analysis, Data and Assumption Register (D2)	WP3	Minute-taker (weeks 1-3)
[Member 3]	Algorithm lead	Task 3 material-movement algorithm, Task 4 assumptions (D3)	WP4	AI/software verification officer (weeks 1-4)
[Member 4]	Block model and validation lead	Task 5 block model, Task 6 validation (D4)	WP2	Meeting chair (weeks 4-6)
[Member 5]	Platform, uncertainty and reproducibility lead	Task 7 uncertainty and sensitivity, Task 8 automated platform, tests, audit trail (D5)	WP1	EDI and professional-practice champion

Roles rotate where indicated, so that leadership and administrative load are shared fairly. The **AI/software verification officer** keeps the AI-use log (D6 appendix) and checks that every AI-assisted code change has a recorded test or hand check. The **EDI and professional-practice champion** leads the ethics/equity scenario discussion and checks that the report states its limitations.

2. Communication expectations and channels

Channel	Purpose	Response expectation
Team chat (e.g. Teams/Discord channel)	Day-to-day coordination, quick questions	Within 24 h on weekdays
Shared Git repository (this repo)	All code, configuration, data registry, documents; one branch per work package, merged after review	Pull requests reviewed within 48 h
Weekly meeting (in person or video, 60 min)	Progress, decisions, blockers	Attendance expected; notify the chair 24 h ahead if absent
Shared drive	Large binaries (presentation recordings, scanned signatures)	
E-mail to the instructor/TA	Only via the project lead, after team agreement	

Decisions made in chat are copied into the decision log (Section 8) by the person who proposed them.

3. Decision-making process

1. **Technical decisions inside one work package:** the owner decides after consulting the reviewer and records the decision in the decision log with its justification.
2. **Decisions affecting more than one work package or the final results** (for example survey timing, density basis, the algorithm form, block size): discussed at the weekly meeting. We aim for **consensus**. If no consensus is reached after a 10-minute structured discussion (each member states a position without interruption), we take a **majority vote** of members present (quorum of 4). The dissenting view is recorded in the decision log.
3. **Engineering-judgement items** (exclusions, corrections, imputations) need a register entry: assumption, need, justification, consequence and treatment (Task 4 format). The reviewer approves it before the change is merged.

4. Procedure for resolving disagreements

1. Raise the issue directly and respectfully with the person concerned within 48 h. Focus on the work, not the person.
2. If unresolved, bring it to the weekly meeting as an agenda item. The chair facilitates, and each side presents evidence (data, test results, references).
3. If still unresolved, a vote decides, as in Section 3, and the minority position is documented.
4. For persistent problems (missed commitments, conduct), the project lead holds a private conversation and agrees a written recovery plan. If this fails, the team informs the instructor, with a factual summary that the member concerned has seen.

5. Meeting and contribution expectations

- Weekly 60-minute meeting, plus a short stand-up in the lab session. The agenda is posted 24 h before. Minutes are posted within 24 h and record attendance, decisions, actions, owners and due dates.

- Each member is expected to contribute about equal effort (about 6-8 h per week). Workload is reviewed every second meeting. If one person is overloaded, tasks are re-allocated and the change is recorded.
- Commitments are made explicitly ("I will do X by date Y"). If a commitment is at risk, the member tells the team **before** the due date.
- Every member presents part of D7 and must be able to explain any part of the method.

6. Internal technical review and quality control

Item	QC step	Evidence
Code	Peer review of every pull request by the WP reviewer. Automated tests (`python -m pytest`) must pass	PR approval, test log
AI-assisted code or text	Verification by a member other than the one who prompted the AI. Hand calculation or test on a small case	AI-use log entry (D6, Appendix A)
Data cleaning or exclusion	Register entry approved by the reviewer. Original record retained with a flag	`outputs/qaqc/fms_records_with_flags.csv`
Numbers in the report	Each number traced to an output file produced by a logged run (`outputs/audit/run_manifest.json`)	Run ID cited in the report
Final submission	Checklist (course outline, Section 12) ticked by two members independently	Signed checklist

7. Respectful and inclusive participation

- Everyone gets the floor. The chair actively invites quieter members to speak before decisions, and written input (chat or comments) counts the same as speaking.
- Meeting times rotate where needed to accommodate work, caregiving, religious observance and time zones. Accessibility needs are accommodated without requiring disclosure of reasons.
- Feedback is specific, about the work, and given in private when it is critical.
- Technical tasks are assigned by interest and development goals, not by assumptions about who is "good at" coding, writing or presenting. Each member gets at least one opportunity to lead.
- Credit is attributed accurately in the contribution record.
- Harassment or discriminatory behaviour is not tolerated and will be escalated under university policy.

8. Work plan (Fall 2026)

Week (starting)	Milestone	Owner(s)	Deliverable / evidence
Oct 5	Charter signed; repository and environment set up; data inventory	All, M5	D1, `requirements.txt`

Week (starting)	Milestone	Owner(s)	Deliverable / evidence
Oct 12	Task 1 QA/QC complete; survey re-dating and timing decisions logged	M2	QA/QC tables and figures
Oct 19	Task 2 growth and FMS-survey reconciliation; density/swell discussion	M2, M4	Period reconciliation table
Oct 26	Task 3 algorithm prototype; forward physics model; Task 4 register v1	M3	Algorithm notebook, D2 draft
Nov 2	Block model v1; mass/metal balance verified; unit tests	M4, M5	D4 draft, test log
Nov 9	Validation (temporal, twin experiment), Monte Carlo and sensitivity	M4, M3, M5	Validation tables
Nov 16	Platform hardening (update workflow, audit trail); report draft; ethics/equity discussion	M5, M1, all	D5, D6 draft
Nov 23	Final review, presentation rehearsal, submission zip (Thursday Nov 26, 23:59)	All	D1-D9

Buffer: the 48 h before the deadline are reserved for review only, not new analysis.

8.1 Meeting log and contribution record (to be maintained by the team)

Date	Attendees	Key decisions (ID)	Actions (owner, due)	Changes in responsibility

8.2 Technical decision log (decisions taken in the analysis - assign the proposer and approver)

ID	Decision	Rationale (see D2/D3)	Proposed by	Approved by / date
DEC-01	Survey files `Topo_250328` / `Topo_250329` re-dated to 2026-03-28/29	QA-S02: FMS dates, face progression, layer naming		
DEC-02	~~Survey surfaces represent the end of the day (23:59)~~ **Superseded by DEC-11**	The 21:00-23:59 search range ended at its own optimum		
DEC-03	Treat each DXF as a closed change solid (bottom = previous surface)	100% of cells hit by exactly two faces; bottoms match previous tops		
DEC-04	Use FMS dump XY only, not dump Z	Z residual vs surface sd ≈ 2.5 m		
DEC-05	No record deleted; flags + documented treatment	Traceability requirement		

ID	Decision	Rationale (see D2/D3)	Proposed by	Approved by / date
DEC-06	Allocation by entropy-regularised optimal transport (CPST)	Survey-constrained, mass-conserving, mixing parameter explicit		
DEC-07	~~Base-case density = FMS density (κ = 1.0); κ tested 0.80-1.05~~ **Superseded by DEC-12**	The range stopped at κ = 0.80 (f_v = 1.25) and never reached the bulking hypothesis (κ ≈ 0.68)		
DEC-08	Block size 5 × 5 × 2 m	Survey/GPS resolution, reclaim equipment, sensitivity tested		
DEC-10	~~Impute X/Y of 14 records with an identical default position (A-32)~~ **Superseded by DEC-13**	QA-F11: physically impossible location		
DEC-09	Legacy material reported separately with assumed grade, never mixed into “measured” totals	Professional accountability (Section 6.2 of brief)		
DEC-11	Survey labelled D closes its interval at D+1 07:00 (cut-off 31 h after 00:00 of D)	0-36 h scan: plateau 30-31.5 h (r 0.992 vs 0.961 at midnight); no dumps 06:00-07:30 (shift change) (A-05)		
DEC-12	In-pile volume = T × f_v / Dens with f_v calibrated on consistent intervals (1.47); base case payloads as recorded (f_pay = 1, i.e. bulking); f_pay 1.00-1.19 in the Monte Carlo; missing-load interpretation reported as a scenario outside the ranges	Hold-out error 3.5% vs 32% for f_v = 1 (A-06)		
DEC-13	Dump location of LOC_DEFAULT records treated as unknown (uniform push cost) instead of imputed	An imputed centroid is still an invented position (A-32)		
DEC-14	A-13b (inferred toe fill) off in the base case; kept as a sensitivity	With f_v calibrated, complete days have no room for 7,200 m³ of unrecorded fill		
DEC-15	Headline inventory = record-period material (FMS-tracked + unattributed), tied to surveyed solids; pre-record core and whole-pile total reported second	65% of the whole-pile tonnes rest on inferred geometry and assumed grade		

ID	Decision	Rationale (see D2/D3)	Proposed by	Approved by / date
DEC-16	Merge of the independent implementation B: adopt its timing scan, volume-factor calibration and hold-out, unknown-location treatment, hashed raw store with interventions, QA rule catalogue and test philosophy; keep A's CPST engine, twin tests, 5 × 5 × 2 m blocks and built deliverables	Cross-implementation review (Appendix A, D6 Section 9.8)		
DEC-17	Keep the slope-aware push cost in the base case, report the flat-cost ablation, and sample both cost forms in the Monte Carlo	Twins: the push cost helps only when the truth follows the same physics (D6 Section 6)		

9. Agreement

By signing, each member agrees to follow this charter and to raise concerns early.

Name	Signature	Date
[Member 1]		
[Member 2]		
[Member 3]		
[Member 4]		
[Member 5]		